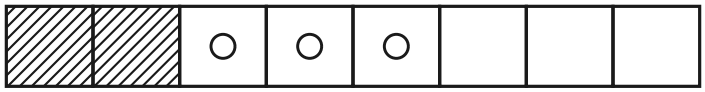
Adding and Subtracting Fractions

Same size pieces: add or take away the tops.

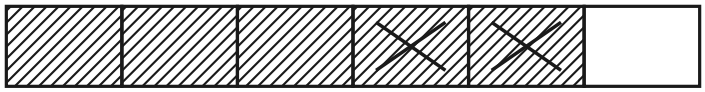
2/8 + 3/8: count the eighths



$$\frac{2}{8} + \frac{3}{8} = \frac{5}{8}$$

2 eighths and 3 more eighths make 5 eighths. The bottom number stays.

5/6 – 2/6: take away sixths



$$\frac{5}{6} - \frac{2}{6} = \frac{3}{6}$$

Split a fraction into parts

$$\frac{3}{5} = \frac{1}{5} + \frac{1}{5} + \frac{1}{5}$$

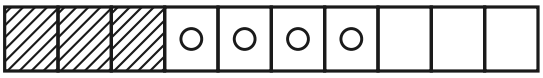
$$\frac{3}{5} = \frac{1}{5} + \frac{2}{5}$$

Any way you split the pieces, they add up to the same amount.

Fraction buddies make one whole

$$\frac{3}{8} + \frac{5}{8} = \frac{8}{8} = 1$$

Your turn!



$$\frac{3}{10} + \frac{4}{10} = \frac{\boxed{7}}{10}$$

For teachers

Goal: add and subtract fractions with the same bottom number, and split a fraction into parts (4.NF.3a, b, d). Say "What size are the pieces? How many pieces now?" Never add the bottom numbers.

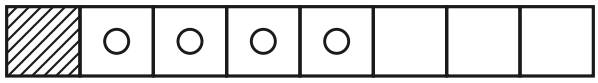
Adding and Subtracting Fractions: Practice

The bottom number tells the size of the pieces. It stays the same.

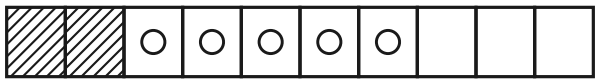
Part 1 Add with strips Shaded pieces, then the dotted pieces.

1) 

$\frac{3}{6} + \frac{2}{6} = \frac{\boxed{5}}{6}$

2) 

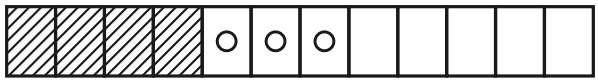
$\frac{1}{8} + \frac{4}{8} = \frac{\boxed{5}}{8}$

3) 

$\frac{2}{10} + \frac{5}{10} = \frac{\boxed{7}}{10}$

4) 

$\frac{1}{3} + \frac{1}{3} = \frac{\boxed{2}}{3}$

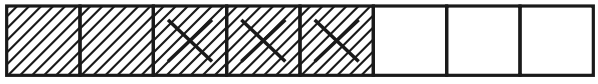
5) 

$\frac{4}{12} + \frac{3}{12} = \frac{\boxed{7}}{12}$

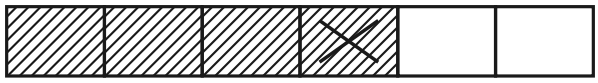
6) 

$\frac{2}{5} + \frac{2}{5} = \frac{\boxed{4}}{5}$

Part 2 Take away with strips Cross out the pieces you take away.

7) 

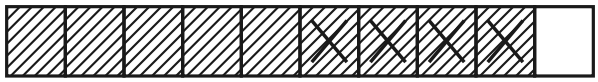
$\frac{5}{8} - \frac{3}{8} = \frac{\boxed{2}}{8}$

8) 

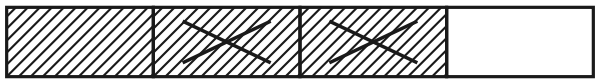
$\frac{4}{6} - \frac{1}{6} = \frac{\boxed{3}}{6}$

Adding and Subtracting Fractions: Practice (continued)

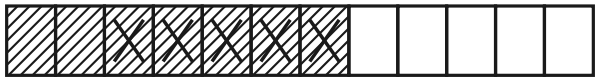
Part 2 Take away with strips (continued)

9) 

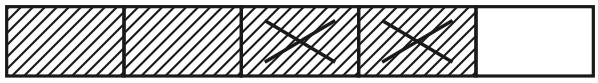
$\frac{9}{10} - \frac{4}{10} = \boxed{\frac{5}{10}}$

10) 

$\frac{3}{4} - \frac{2}{4} = \boxed{\frac{1}{4}}$

11) 

$\frac{7}{12} - \frac{5}{12} = \boxed{\frac{2}{12}}$

12) 

$\frac{4}{5} - \frac{2}{5} = \boxed{\frac{2}{5}}$

Part 3 Split it up Find the missing part.

13) $\frac{5}{8} = \frac{2}{8} + \boxed{\frac{3}{8}}$

14) $\frac{4}{6} = \frac{1}{6} + \boxed{\frac{3}{6}}$

15) $\frac{7}{10} = \frac{3}{10} + \boxed{\frac{4}{10}}$

16) $\frac{3}{4} = \frac{2}{4} + \boxed{\frac{1}{4}}$

17) $\frac{9}{12} = \frac{4}{12} + \boxed{\frac{5}{12}}$

18) $\frac{4}{5} = \frac{1}{5} + \boxed{\frac{3}{5}}$

19) $\frac{6}{8} = \frac{1}{8} + \boxed{\frac{5}{8}}$

20) $\frac{8}{10} = \frac{6}{10} + \boxed{\frac{2}{10}}$

Adding and Subtracting Fractions: Practice (continued)

Part 3 Split it up (continued)

21) $\frac{5}{6} = \frac{3}{6} + \frac{\boxed{2}}{6}$

22) $\frac{11}{12} = \frac{5}{12} + \frac{\boxed{6}}{12}$

Part 4 Add No help this time.

23) $\frac{3}{8} + \frac{4}{8} = \frac{\boxed{7}}{8}$

24) $\frac{2}{5} + \frac{3}{5} = \frac{\boxed{5}}{5}$

25) $\frac{5}{10} + \frac{2}{10} = \frac{\boxed{7}}{10}$

26) $\frac{1}{6} + \frac{3}{6} = \frac{\boxed{4}}{6}$

27) $\frac{7}{12} + \frac{4}{12} = \frac{\boxed{11}}{12}$

28) $\frac{1}{4} + \frac{2}{4} = \frac{\boxed{3}}{4}$

29) $\frac{4}{10} + \frac{5}{10} = \frac{\boxed{9}}{10}$

30) $\frac{2}{3} + \frac{1}{3} = \frac{\boxed{3}}{3}$

31) $\frac{3}{8} + \frac{3}{8} = \frac{\boxed{6}}{8}$

Adding and Subtracting Fractions: Practice (continued)

Part 4 Add (continued)

32) $\frac{5}{12} + \frac{6}{12} = \frac{11}{12}$

33) $\frac{1}{6} + \frac{4}{6} = \frac{5}{6}$

34) $\frac{2}{10} + \frac{4}{10} = \frac{6}{10}$

35) $\frac{1}{5} + \frac{1}{5} = \frac{2}{5}$

36) $\frac{6}{8} + \frac{1}{8} = \frac{7}{8}$

37) $\frac{3}{12} + \frac{8}{12} = \frac{11}{12}$

Part 5 Take away No help this time.

38) $\frac{7}{8} - \frac{3}{8} = \frac{4}{8}$

39) $\frac{4}{5} - \frac{2}{5} = \frac{2}{5}$

40) $\frac{9}{10} - \frac{5}{10} = \frac{4}{10}$

41) $\frac{5}{6} - \frac{1}{6} = \frac{4}{6}$

42) $\frac{11}{12} - \frac{6}{12} = \frac{5}{12}$

43) $\frac{3}{4} - \frac{1}{4} = \frac{2}{4}$

Adding and Subtracting Fractions: Practice (continued)

Part 5 Take away (continued)

44) $\frac{6}{8} - \frac{4}{8} = \frac{\boxed{2}}{8}$

45) $\frac{8}{10} - \frac{3}{10} = \frac{\boxed{5}}{10}$

46) $\frac{2}{3} - \frac{1}{3} = \frac{\boxed{1}}{3}$

47) $\frac{10}{12} - \frac{7}{12} = \frac{\boxed{3}}{12}$

48) $\frac{6}{6} - \frac{2}{6} = \frac{\boxed{4}}{6}$

49) $\frac{3}{5} - \frac{2}{5} = \frac{\boxed{1}}{5}$

Part 6 Fraction buddies No help this time. Together they make 1 whole.

50) $\frac{5}{8} + \frac{\boxed{3}}{8} = 1$

51) $\frac{2}{5} + \frac{\boxed{3}}{5} = 1$

52) $\frac{7}{10} + \frac{\boxed{3}}{10} = 1$

53) $\frac{1}{6} + \frac{\boxed{5}}{6} = 1$

54) $\frac{5}{12} + \frac{\boxed{7}}{12} = 1$

55) $\frac{1}{4} + \frac{\boxed{3}}{4} = 1$

Adding and Subtracting Fractions: Practice (continued)

Part 6 Fraction buddies (continued)

56) $\frac{4}{10} + \frac{\boxed{6}}{10} = 1$

57) $\frac{2}{3} + \frac{\boxed{1}}{3} = 1$

58) $\frac{7}{8} + \frac{\boxed{1}}{8} = 1$

Part 7 Add

59) $\frac{3}{12} + \frac{5}{12} = \frac{\boxed{8}}{12}$

60) $\frac{6}{10} + \frac{2}{10} = \frac{\boxed{8}}{10}$

61) $\frac{3}{10} + \frac{7}{10} = \frac{\boxed{10}}{10}$

62) $\frac{6}{10} + \frac{1}{10} = \frac{\boxed{7}}{10}$

63) $\frac{7}{10} + \frac{1}{10} = \frac{\boxed{8}}{10}$

64) $\frac{4}{6} + \frac{2}{6} = \frac{\boxed{6}}{6}$

65) $\frac{3}{6} + \frac{3}{6} = \frac{\boxed{6}}{6}$

66) $\frac{1}{10} + \frac{7}{10} = \frac{\boxed{8}}{10}$

67) $\frac{6}{10} + \frac{4}{10} = \frac{\boxed{10}}{10}$

Adding and Subtracting Fractions: Practice (continued)

Part 7 Add (continued)

68) $\frac{1}{5} + \frac{3}{5} = \frac{4}{5}$

69) $\frac{2}{12} + \frac{5}{12} = \frac{7}{12}$

70) $\frac{5}{10} + \frac{1}{10} = \frac{6}{10}$

71) $\frac{2}{6} + \frac{2}{6} = \frac{4}{6}$

72) $\frac{5}{8} + \frac{3}{8} = \frac{8}{8}$

73) $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$

74) $\frac{2}{8} + \frac{4}{8} = \frac{6}{8}$

75) $\frac{1}{8} + \frac{4}{8} = \frac{5}{8}$

76) $\frac{4}{8} + \frac{3}{8} = \frac{7}{8}$

77) $\frac{1}{10} + \frac{5}{10} = \frac{6}{10}$

78) $\frac{3}{5} + \frac{2}{5} = \frac{5}{5}$

79) $\frac{2}{5} + \frac{2}{5} = \frac{4}{5}$

Adding and Subtracting Fractions: Practice (continued)

Part 7 Add (continued)

80) $\frac{5}{10} + \frac{5}{10} = \frac{\boxed{10}}{10}$

81) $\frac{2}{6} + \frac{1}{6} = \frac{\boxed{3}}{6}$

82) $\frac{3}{10} + \frac{4}{10} = \frac{\boxed{7}}{10}$

Part 8 Take away

83) $\frac{3}{10} - \frac{2}{10} = \frac{\boxed{1}}{10}$

84) $\frac{4}{6} - \frac{3}{6} = \frac{\boxed{1}}{6}$

85) $\frac{2}{10} - \frac{1}{10} = \frac{\boxed{1}}{10}$

86) $\frac{5}{6} - \frac{2}{6} = \frac{\boxed{3}}{6}$

87) $\frac{2}{5} - \frac{1}{5} = \frac{\boxed{1}}{5}$

88) $\frac{4}{6} - \frac{2}{6} = \frac{\boxed{2}}{6}$

89) $\frac{5}{6} - \frac{4}{6} = \frac{\boxed{1}}{6}$

90) $\frac{6}{8} - \frac{5}{8} = \frac{\boxed{1}}{8}$

91) $\frac{3}{6} - \frac{2}{6} = \frac{\boxed{1}}{6}$

Adding and Subtracting Fractions: Practice (continued)

Part 8 Take away (continued)

92) $\frac{5}{8} - \frac{2}{8} = \frac{3}{8}$

93) $\frac{3}{6} - \frac{1}{6} = \frac{2}{6}$

94) $\frac{7}{8} - \frac{5}{8} = \frac{2}{8}$

95) $\frac{5}{8} - \frac{1}{8} = \frac{4}{8}$

96) $\frac{7}{10} - \frac{1}{10} = \frac{6}{10}$

97) $\frac{8}{12} - \frac{7}{12} = \frac{1}{12}$

98) $\frac{4}{5} - \frac{3}{5} = \frac{1}{5}$

99) $\frac{8}{10} - \frac{4}{10} = \frac{4}{10}$

100) $\frac{4}{8} - \frac{2}{8} = \frac{2}{8}$

Part 9 Remember? Problems from earlier packets.

101) $\frac{3}{4} = \frac{75}{100}$

102) $\frac{2}{3} < \frac{7}{8}$

Adding and Subtracting Fractions: Practice (continued)

Part 9 Remember? (continued)

103) $64 \times 15 =$ **960**

104) $1{,}035 \div 5 =$ **207**

105) $\frac{4}{5}$ of $40 =$ **32**

106) $2{,}999 + 1{,}436 =$ **4,435**

Part 10 Word problems Draw a strip.

107) Mia ate 2/8 of a pizza. Ben ate 3/8 of it. How much of the pizza did they eat?

Number sentence **2/8 + 3/8 = 5/8**

Answer **5/8** of the pizza

Draw or show your work

108) A jug is 7/10 full. Sam drinks 3/10 of the jug. How full is it now?

Number sentence **7/10 − 3/10 = 4/10**

Answer **4/10** full

Draw or show your work

109) Leo walked 1/4 mile to the park and 2/4 mile to school. How far did he walk?

Number sentence **1/4 + 2/4 = 3/4**

Answer **3/4** mile

Draw or show your work

110) A ribbon is 5/6 of a yard. Kim cuts off 4/6 of a yard. How much is left?

Number sentence **5/6 − 4/6 = 1/6**

Answer **1/6** of a yard

Draw or show your work

Adding and Subtracting Fractions: Practice (continued)

Part 10 Word problems (continued)

111) Ana painted 5/12 of a wall. How much more must she paint to finish the wall?

Number sentence **5/12 + 7/12 = 1**

Answer **7/12** of the wall

Draw or show your work

112) Challenge: Jo read 2/10 of a book on Monday and 5/10 on Tuesday. How much is left to read?

Number sentence **2/10 + 5/10 = 7/10, 1 − 7/10 = 3/10**

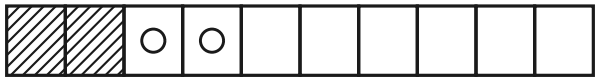
Answer **3/10** of the book

Draw or show your work

Check-In: Adding and Subtracting Fractions

Show what you know. No help this time. **Mastery: 8 of 9 (80%)**

Part 1 Add with strips

1) 

$$\frac{2}{10} + \frac{2}{10} = \boxed{4}{10}$$

Part 2 Split it up

2) $\frac{6}{8} = \frac{3}{8} + \boxed{3}{8}$

3) $\frac{4}{6} = \frac{2}{6} + \boxed{2}{6}$

Part 3 Add

4) $\frac{2}{10} + \frac{5}{10} = \boxed{7}{10}$

5) $\frac{9}{12} + \frac{2}{12} = \boxed{11}{12}$

6) $\frac{4}{8} + \frac{1}{8} = \boxed{5}{8}$

Part 4 Take away

7) $\frac{3}{4} - \frac{2}{4} = \boxed{1}{4}$

8) $\frac{8}{12} - \frac{3}{12} = \boxed{5}{12}$

9) $\frac{2}{4} - \frac{1}{4} = \boxed{1}{4}$